Shoulder MRI — 30 Sample Cases

A worked teaching file of the 30 most common shoulder MRI diagnoses, with realistic clinical vignettes, filled-in findings, and concise impressions.

Rotator Cuff

Case 1 — Supraspinatus tendinosis

Clinical: 47-year-old male recreational tennis player with insidious anterolateral shoulder pain and painful arc over 4 months.

> **F:** On the oblique coronal and sagittal **fat-suppressed T2/PD** sequences, the **supraspinatus** tendon shows intermediate-grade intratendinous signal hyperintensity within the **critical zone** approximately 12 mm from the footprint, without fluid-bright signal and without a discrete fiber gap. The tendon is mildly thickened. The articular and bursal surfaces remain intact with a continuous footprint at the greater tuberosity. No subacromial-subdeltoid fluid. Mild **type II (curved) acromion**. Muscle bulk is preserved with **Goutallier 0**. Normal AC joint and glenohumeral cartilage.

> **I:** Supraspinatus tendinosis without tear. No full- or partial-thickness defect.

Case 2 — Articular-side partial-thickness tear (PASTA)

Clinical: 34-year-old female competitive swimmer with anterior pain on overhead reach and weakness in abduction.

> **F:** Oblique coronal **fat-suppressed T2** images demonstrate a fluid-signal cleft along the **articular (undersurface)** fibers of the **distal supraspinatus** at the footprint, measuring ~5 mm in AP depth and involving roughly 40% of the ~11 mm tendon thickness, consistent with a **partial articular-sided tendon avulsion (PASTA)**. The bursal-side fibers and footprint cortex are intact. Minimal reactive marrow edema at the greater tuberosity. No tendon retraction; muscle bulk normal (**Goutallier 0**). Trace glenohumeral joint fluid.

> **I:** Articular-side partial-thickness supraspinatus tear (PASTA), grade ~40% (intermediate-depth). No full-thickness component.

Case 3 — Bursal-side partial-thickness tear

Clinical: 58-year-old male warehouse worker with chronic impingement-type pain worse with overhead lifting.

> **F:** Fluid-bright signal involves the **bursal-surface** fibers of the **supraspinatus** near the footprint, with focal fiber irregularity over a ~7 mm AP segment and depth involving ~30% of tendon thickness; the articular surface remains continuous. Associated **subacromial-subdeltoid bursal fluid** and reactive bursal thickening. **Type III (hooked) acromion** with a small enthesophyte and mild downsloping. Greater tuberosity cortical irregularity. **Goutallier 0–1**.

> **I:** Bursal-side partial-thickness supraspinatus tear in the setting of subacromial impingement.

Case 4 — Full-thickness supraspinatus tear with retraction and fatty atrophy

Clinical: 64-year-old male, fall onto outstretched arm 6 weeks ago, now with weakness and positive drop-arm sign.

> **F:** A **full-thickness tear** of the **supraspinatus** is present with a complete fluid-filled gap; the AP (coronal) tear dimension is ~18 mm and the retracted tendon stump lies at the level of the glenoid rim (**Patte stage 2**, **medial retraction ~22 mm**). The footprint is denuded. **Subacromial-subdeltoid and glenohumeral fluid**. The supraspinatus muscle belly shows **Goutallier grade 2** fatty infiltration with mild atrophy on the parasagittal **T1** images (positive tangent sign borderline). Infraspinatus intact.

> **I:** Retracted full-thickness supraspinatus tear (~18 mm, Patte 2) with Goutallier 2 fatty atrophy. Repairable but at-risk myotendinous quality.

Case 5 — Massive cuff tear with superior humeral migration

Clinical: 73-year-old female with chronic shoulder pain, pseudoparalysis, and inability to actively elevate the arm.

> **F:** Confluent **full-thickness tears** involve the **supraspinatus and infraspinatus** (≥2 tendons, AP width >5 cm), with extensive medial retraction to the glenoid and beyond (**Patte 3**). The **acromiohumeral interval is narrowed to 4 mm** with superior subluxation of the humeral head and remodeling/concavity of the undersurface of the acromion (early **cuff-tear arthropathy / "femoralization"**). **Goutallier grade 3–4** fatty atrophy of supraspinatus and infraspinatus with positive **tangent sign**. Biceps long head is medially subluxed.

> **I:** Massive (2-tendon) retracted rotator cuff tear with superior humeral migration and advanced fatty atrophy — features of early cuff-tear arthropathy, likely irreparable.

Case 6 — Subscapularis tear with biceps medial subluxation

Clinical: 52-year-old male, forced external rotation/abduction injury, with weakness on lift-off and belly-press.

> **F:** The **upper subscapularis** demonstrates a full-thickness tear of the superior 1/3 with a ~9 mm fluid gap and medial fiber retraction; the lower fibers remain attached at the lesser tuberosity. The **medial sling (coracohumeral/superior glenohumeral ligaments)** is disrupted, and the **long head of biceps tendon is medially subluxed** out of the bicipital groove, lying over the torn subscapularis. Mild reactive marrow edema at the lesser tuberosity. **Goutallier 1** of subscapularis. Supraspinatus intact.

> **I:** Partial (upper-third) subscapularis tear with medial sling disruption and medial subluxation of the long head of biceps.

Case 7 — Infraspinatus tear

Clinical: 60-year-old male laborer with posterolateral pain and external-rotation weakness.

> **F:** Oblique coronal and sagittal **fat-suppressed T2** images show a high-grade partial- to small full-thickness tear of the **infraspinatus** at its footprint on the posterior facet of the greater tuberosity, with a ~6 mm fluid-signal defect and adjacent tendinosis. Posterosuperior footprint marrow edema. Supraspinatus shows coexistent tendinosis without full-thickness tear. **Goutallier 1** of infraspinatus; teres minor preserved. Mild posterior capsular fluid.

> **I:** High-grade partial-thickness infraspinatus tear with adjacent supraspinatus tendinosis.

Case 8 — Calcific tendinitis

Clinical: 45-year-old female with acute, severe, non-traumatic shoulder pain and markedly restricted motion.

> **F:** A rounded **hypointense (low signal on all sequences) calcific deposit** measuring ~12 mm is seen within the **supraspinatus** tendon near the footprint, with surrounding **fat-suppressed T2** hyperintense edema in the tendon and adjacent **subacromial-subdeltoid bursa** indicating the resorptive/inflammatory phase. No full-thickness tendon defect. Faint blooming is suggested; correlation with radiographs confirms amorphous calcific density. No osseous erosion.

> **I:** Hydroxyapatite calcific tendinitis of the supraspinatus in the resorptive phase with reactive bursitis.

Biceps & Labrum

Case 9 — Long head of biceps tendinosis

Clinical: 49-year-old female with anterior shoulder pain localized to the bicipital groove, tender to palpation.

> **F:** The **long head of the biceps (LHB)** tendon within the bicipital groove is thickened with intermediate intrasubstance signal on **fat-suppressed T2** but no fiber discontinuity. A small amount of fluid surrounds the tendon in the groove (tenosynovial fluid, communicating with the joint) without true effusion-out-of-proportion. The tendon remains seated in the groove; the transverse ligament is intact. No SLAP extension. Rotator cuff intact.

> **I:** Long head of biceps tendinosis with mild tenosynovial fluid. No tear or dislocation.

Case 10 — LHB tear and dislocation

Clinical: 66-year-old male with a "pop" while lifting, anterior ecchymosis and a "Popeye" deformity developing.

> **F:** The **bicipital groove is empty** of tendon; the **long head of biceps** is **medially dislocated**, displaced over a torn **subscapularis** insertion, with the medial sling disrupted. More distally the tendon shows discontinuity/retraction compatible with a high-grade/complete tear and distal bunching of the muscle. Fluid tracks along the groove. Associated upper subscapularis tearing. Supraspinatus tendinosis.

> **I:** Long head of biceps dislocation with high-grade tear, associated with subscapularis insertional tear and medial sling disruption.

Case 11 — SLAP II tear

Clinical: 27-year-old male overhead baseball pitcher with deep posterior shoulder pain and loss of velocity; positive O'Brien.

> **F:** On **fat-suppressed T2** and the **ABER/MR arthrogram** images, contrast/fluid signal undermines the **superior labrum** and extends both anterior and posterior to the biceps anchor, separating the labral-biceps complex from the superior glenoid (**SLAP type II**). The signal is laterally curved and extends >2 mm medial to the glenoid rim, distinguishing it from a sublabral recess. The biceps anchor is detached. No paralabral cyst. Rotator cuff intact.

> **I:** SLAP type II tear involving the biceps anchor (anteroposterior).

Case 12 — Bankart lesion (anteroinferior labrum)

Clinical: 21-year-old male rugby player after a traumatic anterior glenohumeral dislocation, now with apprehension.

> **F:** **MR arthrogram** shows detachment/tear of the **anteroinferior labrum** at the 3–6 o'clock position with contrast insinuating between labrum and glenoid; the **scapular periosteum is disrupted** (classic soft-tissue **Bankart**). Associated impaction deformity of the posterosuperolateral humeral head (see Hill-Sachs). The anterior band of the **inferior glenohumeral ligament (IGHL)** complex is attached to the avulsed labral fragment. No osseous glenoid fragment on the GRE images.

> **I:** Soft-tissue Bankart lesion (anteroinferior labral tear with periosteal disruption) following anterior dislocation.

Case 13 — Bony Bankart

Clinical: 31-year-old male with recurrent anterior instability after multiple dislocations.

> **F:** There is an **osseous Bankart** with a displaced bony fragment from the **anteroinferior glenoid rim** and a corresponding marginal defect; estimated glenoid bone loss ~18% of the inferior glenoid diameter on the en-face sagittal reformat. The attached anteroinferior labrum/IGHL is avulsed with the fragment. Marrow edema along the fracture margin indicates relative acuity. Engaging Hill-Sachs not demonstrated at neutral.

> **I:** Bony Bankart lesion with ~18% anteroinferior glenoid bone loss — relevant to surgical planning (approaching critical bone loss).

Case 14 — Hill-Sachs lesion

Clinical: 24-year-old female after first-time anterior dislocation, reduced in the ED.

> **F:** A wedge-shaped impaction fracture of the **posterosuperolateral humeral head** is seen on the superior axial sections (the **Hill-Sachs lesion**), measuring ~14 mm in width and ~7 mm in depth, with subchondral marrow edema indicating acuity. The defect lies above the level of the coracoid. Accompanying anteroinferior labral (Bankart) injury. No loose body.

> **I:** Acute Hill-Sachs impaction fracture with associated anteroinferior labral injury (anterior instability complex).

Case 15 — ALPSA / Perthes lesion

Clinical: 29-year-old male with chronic anterior instability and recurrent subluxations after a remote dislocation.

> **F:** **MR arthrogram** demonstrates an **anterior labroligamentous periosteal sleeve avulsion (ALPSA)**: the anteroinferior labrum is avulsed but the scapular periosteum remains **intact and stripped/displaced medially**, with the labroligamentous complex rolled medially and resynovialized/blunted along the glenoid neck. (In the **Perthes** variant the medially intact periosteum keeps the nondisplaced labrum reduced, requiring ABER views to show contrast undercutting.) No osseous fragment.

> **I:** ALPSA lesion — medially displaced, healed-in anteroinferior labrum with intact periosteum; chronic anterior instability.

Case 16 — Posterior labral tear / reverse Bankart

Clinical: 26-year-old male lineman with posterior shoulder pain after a posteriorly directed load; sense of instability.

> **F:** Axial **MR arthrogram** images show a tear/detachment of the **posteroinferior labrum** (7–9 o'clock) with contrast undermining the labrum (**reverse Bankart**); the posterior band of the IGHL and posterior capsule are involved. A small impaction defect of the anteromedial humeral head (**reverse Hill-Sachs / McLaughlin**) is noted. No osseous posterior glenoid fragment. Posterior capsular redundancy.

> **I:** Posterior labral tear (reverse Bankart) with reverse Hill-Sachs — posterior instability complex.

Case 17 — Paralabral cyst with spinoglenoid notch involvement

Clinical: 38-year-old male volleyball player with vague posterior pain and external-rotation weakness; EMG suggested infraspinatus denervation.

> **F:** A lobulated **T2-hyperintense, T1-hypointense paralabral (ganglion) cyst** measuring ~22 × 12 mm arises from a **posterosuperior labral tear** and extends into the **spinoglenoid notch**, compressing the **suprascapular nerve** distal to the supraspinatus branch. The **infraspinatus** muscle shows **T2 hyperintense edema-like denervation signal** with early fatty change (**Goutallier 1**); supraspinatus spared (lesion distal to its branch). Underlying posterosuperior labral tear confirmed.

> **I:** Spinoglenoid notch paralabral cyst from a posterosuperior labral tear causing suprascapular neuropathy with isolated infraspinatus denervation.

Impingement & AC Joint

Case 18 — Subacromial-subdeltoid bursitis

Clinical: 41-year-old female with lateral shoulder pain and painful overhead activity; no trauma.

> **F:** **Fat-suppressed T2** images show distension of the **subacromial-subdeltoid bursa** with fluid and synovial thickening, the bursa measuring up to ~4 mm in thickness, without an underlying full-thickness cuff defect. Mild supraspinatus tendinosis and a mildly downsloping **type II acromion**. No calcific deposit. Glenohumeral joint and labrum unremarkable.

> **I:** Subacromial-subdeltoid bursitis with mild supraspinatus tendinosis; impingement-pattern changes.

Case 19 — AC joint osteoarthritis with downward osteophyte

Clinical: 56-year-old male manual laborer with focal pain over the top of the shoulder, worse on cross-body adduction.

> **F:** The **acromioclavicular joint** demonstrates capsular hypertrophy, subchondral cystic change and marginal osteophytes, with a **downward-projecting inferior osteophyte** that effaces the subacromial fat and impinges upon the **supraspinatus myotendinous junction**. Reactive marrow edema on both sides of the joint. Underlying supraspinatus bursal-surface fraying without full-thickness tear. Mild SASD bursal fluid.

> **I:** AC joint osteoarthritis with an inferior osteophyte contributing to extrinsic supraspinatus impingement.

Case 20 — Os acromiale

Clinical: 33-year-old male with chronic impingement symptoms refractory to therapy; tenderness over the acromion.

> **F:** A well-corticated **unfused acromial ossification center (os acromiale)** is identified at the **meso-acromion** level on the axial images, with a synchondrosis showing mild **fat-suppressed T2** edema and adjacent subchondral cystic change suggesting instability/motion. The undersurface of the mobile fragment shows a small enthesophyte narrowing the subacromial space. Underlying supraspinatus bursal-side fraying. SASD bursal fluid.

> **I:** Symptomatic (meso-) os acromiale with edema across the synchondrosis contributing to impingement.

Other

Case 21 — Adhesive capsulitis (frozen shoulder)

Clinical: 53-year-old diabetic female with progressive global stiffness and night pain; markedly reduced passive external rotation.

> **F: Fat-suppressed T2/post-contrast** images show thickening and edema/enhancement of the **axillary recess capsule and IGHL** (capsule >4 mm) and soft-tissue thickening/edema in the **rotator interval** enveloping the coracohumeral ligament, with obliteration of the subcoracoid fat triangle. Reduced axillary recess volume. No significant joint effusion. Rotator cuff and labrum intact.

> **I:** Findings characteristic of adhesive capsulitis (frozen shoulder), interval and axillary recess phenotype.

Case 22 — Quadrilateral space syndrome / teres minor denervation

Clinical: 36-year-old male overhead athlete with vague posterior shoulder pain and external-rotation weakness.

> **F:** On parasagittal **T1 and fat-suppressed T2** images the **teres minor** shows isolated **T2 hyperintense denervation edema** with early fatty replacement, while deltoid, infraspinatus and supraspinatus are spared — implicating the **axillary nerve branch within the quadrilateral space**. Mild fibrous/vascular narrowing of the quadrilateral space is suggested; no discrete mass. Rotator cuff tendons intact.

> **I:** Isolated teres minor denervation consistent with quadrilateral space syndrome (axillary nerve, posterior branch).

Case 23 — Greater tuberosity contusion / nondisplaced fracture

Clinical: 59-year-old female after a fall onto the shoulder, point tenderness over the greater tuberosity, intact active motion.

> **F:** A **nondisplaced fracture line** through the **greater tuberosity** is seen as a hypointense line on T1 with surrounding extensive **fat-suppressed T2** marrow edema (bone contusion/trabecular microfracture pattern), without cortical step-off >2 mm or fragment displacement. The overlying **supraspinatus/infraspinatus footprint** fibers are intact (no avulsion). Small reactive joint effusion and SASD fluid.

> **I:** Nondisplaced greater tuberosity fracture with bone marrow edema; rotator cuff insertions intact.

Case 24 — Postoperative rotator cuff retear

Clinical: 62-year-old male, 8 months after arthroscopic supraspinatus repair, with recurrent weakness and pain.

> **F:** At the prior **supraspinatus** repair, susceptibility artifact from suture anchors is noted at the greater tuberosity. There is a recurrent **full-thickness fluid-signal gap** at the repaired footprint measuring ~10 mm AP with retraction of the tendon edge to the level of the humeral articular surface, indicating **retear (Sugaya type IV–V)**. Anchors appear seated without pullout. Progression of **Goutallier to grade 2** fatty atrophy compared with expected. SASD fluid.

> **I:** Recurrent full-thickness supraspinatus tear at the repair site (retear) with interval progression of fatty atrophy.

Case 25 — Internal (posterosuperior) impingement

Clinical: 25-year-old male throwing athlete with posterosuperior pain in the late-cocking (ABER) position; GIRD on exam.

> **F: ABER MR arthrogram** images show contact-pattern changes of **posterosuperior impingement**: undersurface (articular-side) fraying/partial tear of the **posterior supraspinatus/anterior infraspinatus** fibers, fraying of the **posterosuperior labrum**, and a subcortical cyst/edema in the **posterosuperior greater tuberosity/humeral head**. Mild posterior capsular thickening. No full-thickness cuff tear.

> **I:** Posterosuperior (internal) impingement of the throwing shoulder — articular-side cuff and posterosuperior labral changes.

Case 26 — Glenohumeral osteoarthritis

Clinical: 68-year-old female with chronic deep aching pain, stiffness, and crepitus; no instability.

> **F:** Diffuse **glenohumeral cartilage loss** with joint-space narrowing, **marginal osteophytes** (notably an inferior humeral head "goat's beard" osteophyte), subchondral cysts and sclerosis, and small intra-articular **loose bodies** are seen. Mild posterior glenoid wear (Walch B pattern) with slight posterior humeral subluxation. Rotator cuff is intact. Mild synovitis and effusion.

> **I:** Primary glenohumeral osteoarthritis with posterior glenoid wear and loose bodies; cuff intact.

Case 27 — HAGL lesion (humeral avulsion of glenohumeral ligament)

Clinical: 28-year-old male after a hyperabduction/external-rotation injury with anterior instability but no labral findings clinically.

> **F: MR arthrogram** shows extravasation of contrast at the **humeral attachment of the inferior glenohumeral ligament**, with the axillary pouch assuming a **"J-shaped" configuration** rather than the normal U, indicating a **HAGL** lesion. The anteroinferior labrum is intact (distinguishing from Bankart). No osseous fragment (i.e., not a BHAGL). Mild subscapularis strain.

> **I:** HAGL lesion (humeral-sided avulsion of the inferior glenohumeral ligament); an instability cause with an intact labrum.

Case 28 — GLAD lesion (glenolabral articular disruption)

Clinical: 30-year-old male after a forced adduction injury; persistent anterior pain without frank instability.

> **F: MR arthrogram** demonstrates a superficial tear of the **anteroinferior labrum** with an adjacent **articular cartilage flap/defect** of the anteroinferior glenoid (**GLAD**), with a small chondral fragment. The IGHL and periosteum remain intact and the labrum is not displaced, accounting for stability despite the chondral injury. Subchondral marrow edema beneath the chondral defect.

> **I:** GLAD lesion — anteroinferior labral tear with associated glenoid chondral injury; typically stable.

Case 29 — Suprascapular notch ganglion

Clinical: 44-year-old male with deep aching shoulder pain and combined supraspinatus/infraspinatus weakness.

> **F:** A multiloculated **paralabral/ganglion cyst** (~25 × 14 mm) tracks to the **suprascapular notch** proximal to the spinoglenoid notch, compressing the **suprascapular nerve** before its supraspinatus branch. Consequently **both supraspinatus and infraspinatus** show denervation edema-like **T2 hyperintensity** with early fatty change (**Goutallier 1**). Arises from a superior/posterosuperior labral tear. Cuff tendons intact.

> **I:** Suprascapular notch ganglion causing suprascapular neuropathy with combined supraspinatus and infraspinatus denervation.

Case 30 — Cuff-tear arthropathy (Hamada) — chronic massive tear

Clinical: 79-year-old female with long-standing pseudoparalysis, prior known massive cuff tear, now with grinding pain.

> **F:** Chronic **massive full-thickness cuff tear** (supraspinatus + infraspinatus, with partial subscapularis involvement) with **Goutallier grade 4** fatty atrophy and positive tangent sign. The **acromiohumeral interval is obliterated (≤2 mm)** with acetabularization of the acromion and femoralization of the humeral head, glenohumeral cartilage loss and superior glenoid erosion — **cuff-tear arthropathy (Hamada grade 4)**. The deltoid is intact and hypertrophied (relevant to reverse arthroplasty candidacy).

> **I:** End-stage cuff-tear arthropathy (Hamada 4) from a chronic massive irreparable cuff tear; deltoid intact.